FILTER DESIGN TEST CASE

SAW (Surface Acoustic Wave) - Band 25 (1900 MHz) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0061
Revision:	Rev B.9
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Design Center:	Irvine Design Center
Created Date:	2024-12-13
Last Modified:	2025-01-12
Status:	FAIL
Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a SAW (Surface Acoustic Wave) filter targeting Band 25 (1900 MHz) for IoT Narrowband Module. The filter is designed on AlN on Si substrate with a target center frequency of 3232.1 MHz and bandwidth of 245.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: SAW (Surface Acoustic Wave)
- Frequency Band: Band 25 (1900 MHz)
- Application: IoT Narrowband Module
- Substrate: AlN on Si
- Package: Fan-Out WLP
- Design Tool: CST Studio Suite 2024
- Design Center: Irvine Design Center

This specification applies to all variants of the SAW (Surface Acoustic Wave) design targeting Band 25 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 3232.1 MHz
Bandwidth (3 dB): 245.3 MHz
Insertion Loss Target: <= 1.4 dB
Return Loss Target: >= 20.0 dB
Out-of-Band Rejection: >= 26.0 dB
Isolation (Duplexer): >= 19.0 dB
Q Factor: 7248
Temperature Coefficient:-20.8 ppm/°C
Die Size: 2.5 x 1.0 mm
Wafer Size: 8-inch
Number of Resonators: 6
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the SAW (Surface Acoustic Wave) template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3232.1 MHz, BW 245.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9696 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.4 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 368 MHz offset >= 26.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 8.4 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2025-01-12
Devices Tested: 39
Insertion Loss (meas): 1.12 dB
Return Loss (meas): 21.8 dB
Rejection (meas): 28.5 dB
Isolation (meas): 23.8 dB
Q Factor (meas): 7248
Group Delay Variation: 6.7 ns
Power Handling: +28 dBm
Die Yield: 88.4%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The SAW (Surface Acoustic Wave) design demonstrated below-target performance for Band 25 (1900 MHz).
b) Insertion loss met target specification.
c) Rejection at near-band offset requires additional resonator tuning.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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